

# COVID-19 Data Analysis Project

## Problem Statement

The goal of this project is to analyze global COVID-19 data to understand patterns, trends, and relationships between variables such as total cases, deaths, GDP per capita, and human development index. The project focuses on transforming raw data into meaningful insights through data preprocessing, analysis, and visualization.

## Objectives

- Perform data exploration and understanding
- Clean and preprocess the dataset
- Analyze trends across continents
- Create meaningful visualizations
- Generate insights for decision-making

## Solution Approach

1. Imported dataset using Pandas 2. Performed exploratory data analysis 3. Cleaned missing and duplicate values 4. Transformed date and extracted features 5. Aggregated data by continent 6. Created new ratio-based feature 7. Visualized data using plots 8. Exported processed dataset

## Technologies Used

- Python
- Pandas
- Matplotlib
- Seaborn

## Conclusion

The project successfully demonstrates a complete data analysis workflow, helping to understand COVID-19 trends globally using data science techniques.